

21. Perform Sharpening of Image using Laplacian mask implemented with an extension of diagonal neighbors.

1	1	1
1	-8	1
1	1	1

AIM:

Perform Sharpening of Image using Laplacian mask implemented with an extension of diagonal neighbors.

PROGRAM:

```
import cv2
import numpy as np
```

```
img = cv2.imread(r'C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png')
```

```
if img is None:
```

```
    print("Error: Image not found!") else:
```

```
    # Sharpening kernel
```

```
kernel = np.array([
```

```
    [1, 1, 1],
```

```
    [1, -8, 1],
```

```
    [1, 1, 1]
```

```
])
```

```
    # Apply the sharpening filter
```

```
sharpened = cv2.filter2D(img, -1, kernel)
```

```
    # Save the output image    cv2.imwrite(
```

```
r'C:\Users\Susmitha\Downloads\ITA0510 Lab\Sharpened_Image.jpg',
```

```
sharpened
```

```
)
```

```
    # Display the images
```

```
cv2.imshow("Original Image", img)
```

```
cv2.imshow("Sharpened Image", sharpened)
```

```
cv2.waitKey(0)  
cv2.destroyAllWindows()
```

INPUT:



OUTPUT:



RESULT:

Thus the program is executed successfully.